Figures 1 & 2: Birthweight and mother's weight

- Independent variable (X): Mother's weight (lwt), pounds

- Dependent variable (Y): Birthweight (bwt), grams

- β_0 (intercept): 2369.623518

- β_1 (slope): 4.429108

- R^2: 0.034497

- Explanation: The data show a weak positive relationship: for each 1-pound increase in the mother's weight, predicted birthweight increases by about 4.43 grams.

Figures 3 & 4: Housing insecurity and total population

- Independent variable (X): Total population (total_population)

- Dependent variable (Y): Crude prevalence of housing insecurity (housinsecu_crude_prev)

- β_0 (intercept): 10.667735

- β_1 (slope): −0.001975

- R^2: 0.016422

- Explanation: The data show a very weak negative relationship: for each 1-unit increase in total population, predicted housing insecurity prevalence decreases by about 0.001975 units.

Figures 5 & 6: Obesity and mental health disorders

- Independent variable (X): Crude prevalence of obesity (obesity_crude_prev)

- Dependent variable (Y): Crude prevalence of mental health disorders (mhlth_crude_prev)

- β_0 (intercept): 9.382996

- β_1 (slope): 0.206872

- R^2: 0.241763

- Explanation: The data show a positive relationship of moderate strength: for each 1-unit increase in obesity prevalence, predicted mental health disorder prevalence increases by about 0.207 units.